

# Malo Leroy

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## Experience

### Research Scientist Intern | IBM Quantum

2025 - 2026

*Theoretical and experimental research in Quantum Error Mitigation (QEM)*

[Qiskit](#), [SciPy](#), [NumPy](#)

- Conducted research on a novel QEM technique in a mixed approach of quantum computing theory & data science
- Collaborated with IBM researchers to implement and benchmark the technique on real quantum hardware
- Explored circuit synthesis methods based on decision diagrams for dynamical circuits

### Software Engineer | Paris Digital Lab

2025

*Highly selective gap year program developing three 7-weeks innovative projects for companies*

[Rust](#), [C++](#), [Flutter](#), [Docker](#) 🔗

- Programmed a performance-focused, Cloud-native Fully Homomorphic Encryption framework for BPCE Group
- Engineered a secure phone dialer and messaging app with identity verification based on the european eIDAS / EUDI standard

### Quantum Computing Research Intern | INRIA

2023 - 2025

*Part-time intern in INRIA's QuaCS team*

[QASM](#), [C++](#), [GoogleTest](#), [CMake](#) 🔗

- Theorized an innovative data structure to represent quantum states & circuits
- Started writing a scientific paper on the subject, detailing the mathematical background and properties of the data structure
- Created and implemented algorithms for fast classical simulation of quantum circuits

### General Secretary & Head of Training | ViaRézo (IT association)

2023 - 2025

*Technical administrator of France's largest student ISP*

[Kubernetes](#), [Ansible](#), [OpenStack](#), [Ceph](#)

- Operated 400+ Unifi APs and 50+ Juniper switches to provide Internet to 2.5k+ students
- Administered a 200+ virtual machines OpenStack cluster and a 150+ pods Kubernetes cluster
- Held legal responsibility for database administration (13M+ rows), ensuring european and national compliance (GDPR, NIS2)
- Devised and conducted public training sessions and internal workshops on various technical topics (Bash, Docker,  $\text{\LaTeX}$ , Git)

### Network Infrastructure Intern | Paris-Saclay University

2024

- Troubleshooted and secured a university library's network infrastructure

[Cisco IOS](#), [Wireshark](#), [VLANs](#)

## Projects

### C/C++ Unit testing library | Personal project

2022

- Developed a fast and memory-safe library leveraging the preprocessor, with a stack-based approach
- Engineered pipelines for continuous integration and live documentation deployment

[C](#), [GitHub Actions](#) 🔗

### Classical simulation of quantum algorithms | Chateaubriand Highschool

2022

- Created a formal algebra library to enable exact scalar and matrix computations
- Programmed a standalone proof-of-concept implementation of Shor's algorithm on up to 10 qubits

[Python](#) 🔗

### MP3 player and meta-data reader | Personal project

2021

- Implemented a pure-Dart MP3 meta-data reading library directly from the ID3 standard
- Created a multi-platform MP3 player with a custom UI seamlessly running on Android, iOS, Linux and the web

[Flutter](#), [Dart](#) 🔗

## Education

### CentraleSupélec – Paris-Saclay University

2023-2027

*Master of Engineering (MEng), current GPA 3.97*

- CentraleSupélec is an elite French engineering school part of Paris-Saclay University (12th in the Shanghai Ranking)
- **Relevant Coursework:** networks & security, information theory, microarchitecture design & assembly, linear algebra, general algebra, design and verification of critical systems, quantum physics, quantum computing basics

### QMI master's program – Paris-Saclay University

2026-2027

*Master of Science (MSc), no grades yet*

- Focused on theoretical computer science for quantum done in addition to the CentraleSupélec curriculum, as a dual-degree
- **Relevant Coursework:** tensor networks, quantum hardware, quantum information theory, quantum non-locality, quantum error correction, quantum structures, ZX calculus, quantum compilation, quantum machine learning

### Chateaubriand Highschool

2021-2023

*Selective post-secondary school*

- Intensive two-year preparatory program for competitive entrance exams to French Grandes Ecoles, with a focus on mathematics

🌐 **Languages:** French (native), English (fluent), Spanish (conversational), German (basic)